Survey of Computer Science

Overview

The purpose of this course is to introduce students to computer science as a career by exploring topics such as computer systems, impacts of computing, data manipulation using spreadsheets, programming, the internet, and website development. Assessments will be required during most class sessions so students can display their understanding of the concepts. Collaboration with other students may be required for assessments throughout the course.

Objectives

- Identify impacts of computing in terms of creativity and intellectual property
- Perform calculations on binary representation of numbers and convert bases
- Identify types of data and manipulate data using spreadsheet software
- Apply algorithm design and creation techniques
- Write, analyze, and trace code using the Python programming language
- Define the internet and create a webpage using HTML and CSS
- Identify societal impacts of computing and recognize aspects of computer security

Assessment

Students will be assessed using class activities, quizzes, tests, programs, projects, essays, and presentations. Additionally, students will be prepared to take a CIW Site Development Associate Industry Based Credential.

Course Essentials

Equipment	Cost/Unit
Classroom set of computers	$0 if you already have some, $500-600 per computer if you need to Purchase. Chromebooks are acceptable.

Outline:

Unit 1: Foundations of Computer Science	This unit is an overview of careers in computer science including the history of computer science and computing innovations. Each subsequent unit will explore different careers that students could pursue within computer science.
Unit 2: Computing Systems	Students are introduced to the physical components of computing including motherboards and system architecture.
Unit 3: Software, OS & Applications	Students are provided with the basics of the software side of computing by reviewing operating system fundamentals, software development, and cloud computing.
Unit 4: How Computers Represent Information	Students begin to explore data and information by understanding how computers represent information by exploring notional systems and data encoding and compression.
Unit 5: Data & Information	Students get deeper into data and information by understanding databases, data organization, data encryption, data visualization, and using data for decision making.
Unit 6: Cybersecurity & Digital Citizenship	Students explore cybersecurity as a career through discussions about privacy, security, digital citizenship, online ethics, and intellectual property.
Unit 7: Internet & Networking	Students become familiar internet and networking, cloud computing, web protocols, and internet ethics.
Unit 8: HTML & Web Development	Students are introduced to HTML and CSS to design websites while also learning about accessibility.
Unit 9: Computational Thinking & Problem Solving	Students learn the fundamentals of computational thinking through decomposition, pattern recognition, and algorithmic thinking. This sets the stage for all programming languages.
Unit 10: Programming Fundamentals	Students end the course by understanding the career of a programmer. Students are introduced to Python while exploring expressions, statements, operations, and iterations.

SURVEY OF COMPUTER SCIENCE

1. Materials
   Internet access and 1-to-1 computer use daily.
   Chromebooks, Windows PCs, and Mac computers are acceptable for this course.
2. Required software, networking access, and access to LSU resources:
   - Students and teachers must have reliable access to the Louisiana Training Platform (LTP, http://louisianatrainingplatform.com)
   - Course lessons, instructional videos, interactive activities, Python coding practice, assignments, quizzes, and other course resources are delivered through LTP.
   - District information technology departments should configure network filters, firewalls, and device security settings to allow student and teacher devices to access LTP and the resources required for course instruction. This includes linked websites, embedded videos, interactive activities, and online tools. Required resources should be tested before instruction begins to ensure that security settings do not block access or prevent activities from functioning as intended.
   - Students must be able to upload files (presentation, spreadsheet, document) through LTP when required by course assignments.
   - Whitelist (web-based activities)
     - Binary Game (https://learningcontent.cisco.com/games/binary/index.html)
3. Requirements for IBC Alignment: **CIW Site Development Associate**
   For schools who are pursuing the aligned IBC for this course
   a. Schools must have the appropriate testing software installed on the testing computers. For this credential the testing software is through CIW. More information can be found here: https://ciwcertified.com/exam-requirements/
   b. Some industry based credential providers also provide practice tests softwares such a GMetrix which may be installed. Please check with individual schools if they plan to use a practice software.
   c. Schools must also typically register as an approved testing center prior to being allowed to proctor the exams.
   d. Teachers must have earned the IBC prior to students sitting for the exam. Once they have passed the credential, they need to be set up as a proctor in the CIW system which must be done by the CIW team. Please contact your regional CIW representative.
   e. Teachers should report to the LSU Point of Contact their overall pass rates (i.e. 15/20 students passed)
4. Required teacher collaborations
   Teachers will communicate with LSU instructors via email. Teachers may be asked to share student work, course implementation feedback, or other instructional information with their designated LSU Pathways Point-of-Contact.
5. Required administration of course content, pre/post test, and research instruments
   Required course materials, assessments, activities, and instructional resources will be distributed through the Louisiana Training Platform or other LSU-approved systems identified by the course instructor.

6. Course Work
   Teachers must present the course material in sequence or as approved by collaboration with the LSU Pathway Point-of-Contact. Teachers are expected to deliver a minimum of 80% of the course material for course fidelity and for the opportunity for IBC attainment (where applicable).
7. Other
   As this is a project-based learning class, we strongly suggest that each section of the course be limited to a *maximum* of 20 students. The course is dependent on the teacher providing feedback and reviewing student code. The course requires that teachers have adequate time to interact with each student.